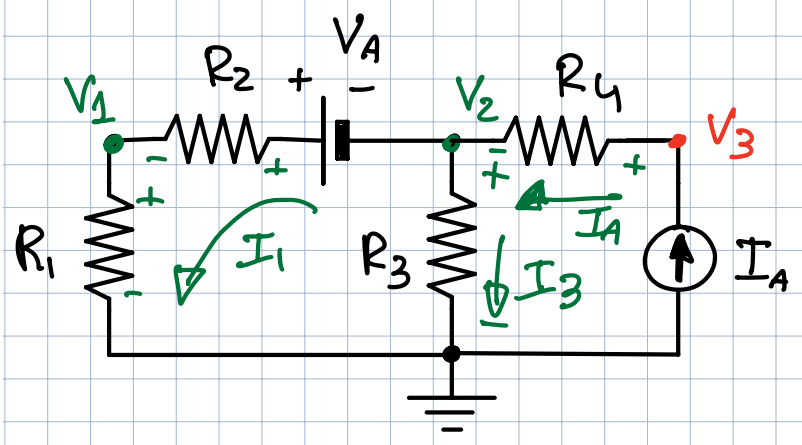
ESERCIZIO 4



$R_1 = 5\,K\Omega$ $\quad V_A = 15\,V$

$R_2 = 10\,K\Omega$ $\quad I_A = 5\,mA$

$R_3 = 5\,K\Omega$

$R_4 = 15\,K\Omega$

RISOLVERE IL CIRCUITO E CALCOLARE V_1 e V_2

$$I_3 = I_A - I_1$$

$$-V_A + (R_2 + R_1) I_1 - R_3 I_3 = 0$$

$$-V_A + (R_2 + R_1) I_1 - R_3 (I_A - I_1) = 0$$

$$\Rightarrow I_1 (R_1 + R_2 + R_3) = V_A + R_3 I_A$$

$$\Rightarrow I_1 = \frac{V_A + R_3 I_A}{R_1 + R_2 + R_3} = \frac{15V + 25V}{20\,K\Omega} = 2\,mA$$

$$I_3 = I_A - I_1 = 5\,mA - 2\,mA = 3\,mA$$

$$\Rightarrow V_2 = R_3 I_3 = 15\,V$$

$$V_2 = (R_1 + R_2) I_1 - V_A = 15\,K\Omega \cdot 2\,mA - 15\,V = 15\,V \quad \checkmark$$

$$V_1 = R_1 I_1 = 5\,K\Omega \cdot 2\,mA = 10\,V$$

$$V_3 = V_2 + R_4 I_A = 15\,V + 15\,K\Omega \cdot 5\,mA = 90\,V$$

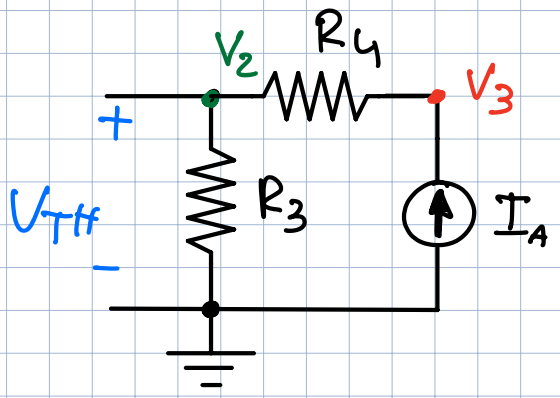
$$V_{R_1R_2} = I_1 (R_1 + R_2) = 30\,V$$

$$P_R = I_1 V_{R_1R_2} + I_3 V_2 + I_A (V_3 - V_2) =$$

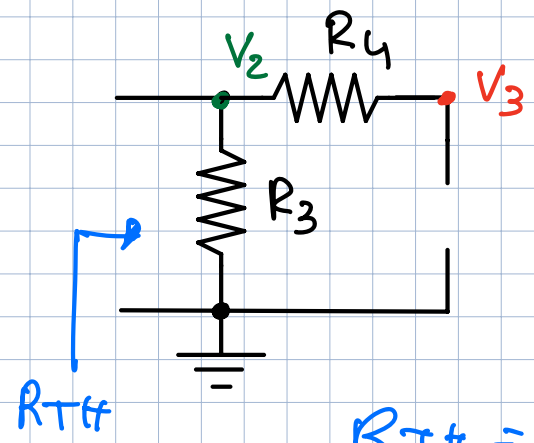
$$= 60\,mW + 45\,mV + 375\,mV = 480\,mV$$

$$P_G = V_A I_1 + I_A V_3 = 15 \times 2\,mA + 90\,V \cdot 5\,mA = 480\,mW$$

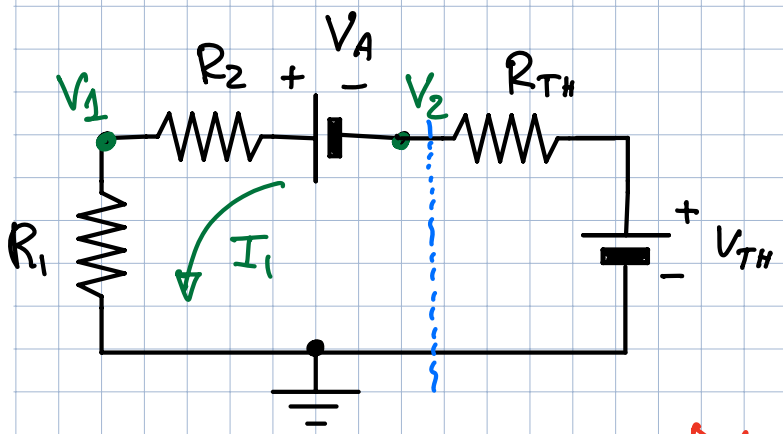
EQUIVALENTE THEVENIN



$$V_{TH} = I_A R_3 = 25\,V$$



$$R_{TH} = R_3$$

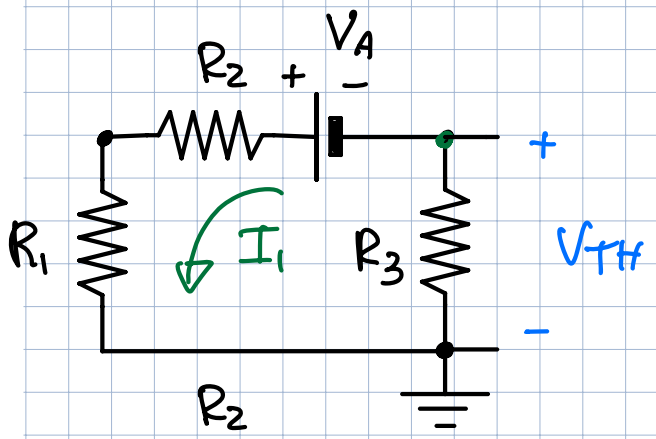


$$I_1 = \frac{V_A + V_{TH}}{R_1 + R_2 + R_3} = \frac{40\,V}{20\,k\Omega} = 2\,mA$$

✓

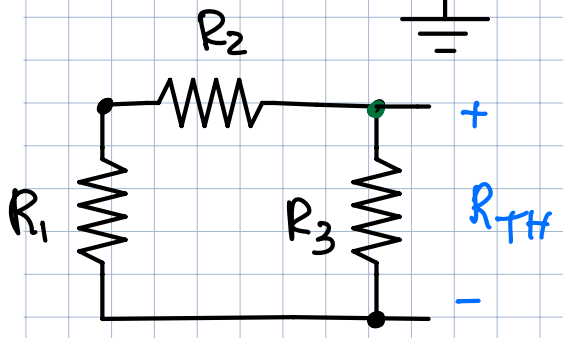
DA QUI USANDO CIRCUITO DI PARTENZA TROVO TUTTO COME PRIMA

EQUIVALENTE THEVENIN 2

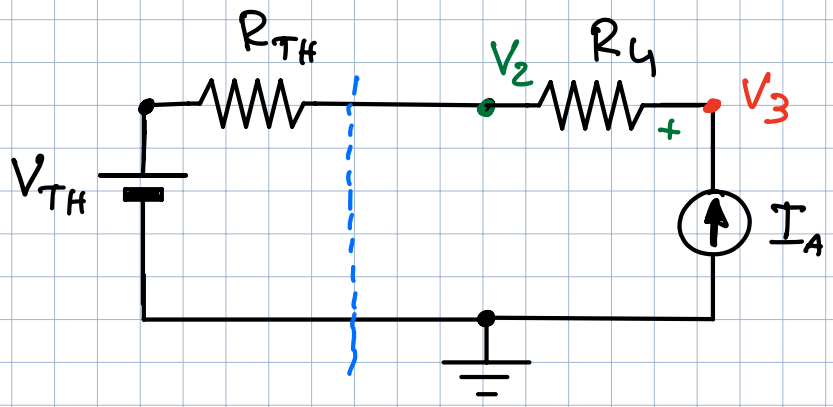


$$I_1 = \frac{V_A}{R_1 + R_2 + R_3} = \frac{15\,V}{20\,k\Omega} = \frac{3}{4}\,mA = 0{,}75\,mA$$

$$V_{TH} = -I_1 R_3 = -\frac{3}{4}\,mA \cdot 5\,k\Omega = -\frac{15}{4}\,V = -3{,}75\,V$$



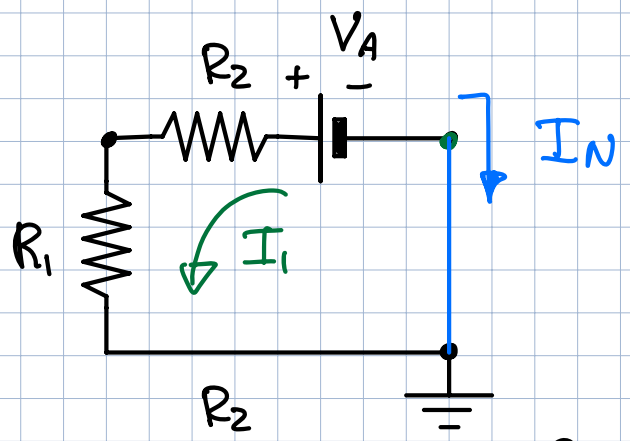
$$R_{TH} = R_3 // (R_1 + R_2) = \frac{5 \cdot 15}{20}\,k\Omega = \frac{75}{20}\,k\Omega = 3{,}75\,k\Omega$$



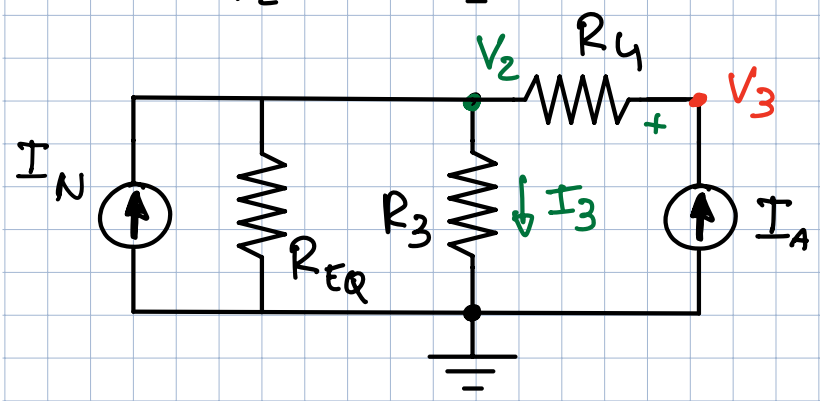
$$V_2 = V_{TH} + R_{TH} I_A$$
$$= -3{,}75V + 3{,}75k\Omega \cdot 5mA$$
$$= 15V \quad ✓$$
$$V_3 = V_2 + R_4 I_A = 90V$$

DA QUI USANDO CIRCUITO DI PARTENZA TROVO TUTTO COME PRIMA

EQUIVALENTE NORTON



$$I_N = -\frac{V_A}{R_1+R_2} = -1mA$$
$$R_{EQ} = R_1 + R_2 = 15k\Omega$$



$$V_2 = (I_A + I_N)\ R_{EQ} \| R_3$$
$$= 4mA \cdot \frac{15 \cdot 5}{20} k\Omega$$
$$= \frac{75}{5} V = 15V \quad ✓$$

$$V_3 = V_2 + I_A R_4 = 15V + 5mA \cdot 15k\Omega = 90V$$
$$I_3 = \frac{V_2}{R_3} = \frac{15V}{5k\Omega} = 3mA$$

DA QUI USANDO CIRCUITO DI PARTENZA TROVO TUTTO COME PRIMA